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Roll No

CE-6003-CBGS

B.E. VI Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Geotechnical Engineering - I

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) The total unit weight of the glacial outwash soil is 16 kN/m^3 .
The specific gravity of soil particles of the soil is 2.67.
The water content of the soil is 17% calculate: 7
i) Dry unit weight
ii) Porosity
iii) Void Ratio
iv) Degree of saturation
b) Define the following terms: 7
i) Coefficient of compressibility
ii) Coefficient of consolidation
iii) Primary consolidation
iv) Over consolidation
2. a) Explain IS soil classification. 7
b) A soil strata consists of 3 layers of thickness 1m, 1.5m and 2m having the coefficient of Permeability of $2 \times 10^{-3} \text{ cm/s}$, $1.5 \times 10^{-3} \text{ cm/s}$ and $3 \times 10^{-3} \text{ cm/s}$ respectively. Estimate the average of coefficient of permeability in the direction. 7
i) Parallel to the bedding plane
ii) Normal to the bedding plane

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3. a) Explain the classification of shear tests based on drainage condition. 7
b) In an unconfined compression test, a sample of sandy clay 8cm long and 4cm in diameter fails under a load of 120N at 10% strain. Compute the shearing resistance taking into account the effect of change in cross-section of the sample. 7
4. a) Compute the intensities of active and passive earth pressure at depth of 8 metres in dry cohesionless sand with an angle of internal friction of 30° and unit weight of 18 kN/m^3 . What will be the intensities of active and passive earth pressure if the water level rises to the ground level? Take saturated unit weight of sand as 22 kN/m^3 . 7
b) Explain the stability of slopes of Earth Dam. 7
5. a) What is Retaining wall? What are its types? Explain any one with proper diagram. 7
b) What is the difference between Rankin's earth pressure theory and Coulomb's earth pressure theory? 7
6. a) Explain the effect of compaction on soil properties. 7
b) Derive an expression to determine coefficient of permeability of soil by laboratory falling head permeability test. 7
7. a) Explain in detail about three clay minerals. 7
b) Explain electrical diffuse double layer and adsorbed water. 7
8. Write short notes 14
 - i) Characteristics of flow net
 - ii) Over consolidated soil
 - iii) Seepage velocity
 - iv) Coefficient of compressibility

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